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MILITARY SPECIFICATION

TANKS, MASTER, X-RAY AND PHOTODOSIMETRIC FILM PROCESSING, MECHANICALLY COOLED, SELF-CONTAINED NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers mechanically cooled, self-contained, developing tanks with a master tank containing 3 insert tanks used for developing X-ray and photodosimetric films.

1.2 Classification. - The film developing tank shall be of the following types, as specified (see 6.1):

Type I - X-Ray film developing tank.
Size 1 - Cabinet with three insert tanks; one insert tank with a capacity of at least 5 gallons and two with a capacity of at least 3 gallons each.
Size 2 - Cabinet with 3 insert tanks; one insert tank with a capacity of at least 10 gallons and two with a capacity of at least 5 gallons each.
Type II - Photo-dosimetric film developing tank.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings and Structural and Special Shaped Sections.
QQ-S-766 - Steel Plates, Sheets and Strip———Corrosion Resisting.
WW-T-799 - Tubing, Copper, Seamless (for Use With Solder -Joint or Flared-Tube Fittings).

MILITARY
MIL-P-3 - Packaging and Packing for Delivery of Valves, Fittings and Flanges (except for Oxygen Systems).
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Production, Electrical Hull and Mechanical Equipment.
MIL-F-1183 - Fittings, Tube Cast, Bronze, Silver Brazing.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment (Naval Shipboard Use).

FSC 6635

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MIL-C-2212 - Controllers, Motor Starters, and Master Switches-Alternating-Current, Naval Shipboard.
MIL-P-12323 - Packaging of Refrigerators.
MIL-T-15005 - Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.
MIL-P-15024 - Plates, Identification——Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action, Bureau Design.
MIL-S-15395 - Silver-Base-Brazing-Alloy.
MIL-I-16022 - Insulation Felt, Thermal, Fibrous Glass, Flexible.
MIL-O-16485 - Ohmeter (Insulation-Resistance-Indicating).
MIL-B-16541 - Bronze, Valve: Castings.
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts).
MIL-F-20042 - Flanges, Pipe, bronze (silver brazing) 50, 100, and 200 P.S.I.W.S.P.
MIL-C-20158 - Copper-Phosphorus-Alloy, Brazing.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Allied Process for Machinery for Ships of the United States Navy.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION

No. MG1 - 1955, Part 7 - Motors for Hermetic Refrigeration Condensing Units.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 E. 44th St. New York 17, N. Y.)

STANDARD FOR SAFETY, CONDENSING UNITS, REFRIGERATION (UNDERWRITERS LABORATORY, INC; UL303)

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York; 207 East Ohio Street, Chicago 11, Illinois; or 500 Sansome Street, San Francisco 11, California.)

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3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with, or the commercial equivalent of, the applicable specifications listed in 2.1 and in conformance with the requirements specified herein.

3.1.1 Corrosion protection. - Corrosion-resisting steel and copper alloys are considered satisfactory corrosion-resisting materials. Where corrosion-resisting steel is specified, it shall conform to class 316 of Specification QQ-S-766. Corrosion-resisting steel, when fabricated by any method that tends to reduce corrosion-resisting properties, shall be normalized to restore those properties before being assembled in any unit.

3.1.1.1 Parts other than corrosion-resisting materials shall be protected against corrosion with chemicals, electrolytic processes, plating, or suitable paints and enamels. The following methods, while not restrictive, shall be considered as a satisfactory corrosion protection when properly applied:

(a) Hot-dipped galvanized.
(b) Hot phosphate process with rust-inhibiting paint.
(c) Electroplated cadmium followed by dichromate, or hot-phosphate treatment or water-dip lacquer.

3.1.1.2 All bolts, nuts, studs, pins, screws and such other fastenings or fittings as may be used shall be of a corrosion-resisting material, or of a material treated in a suitable manner to render it adequately resistant to corrosion. All screws exposed in final assembly shall be corrosion-resisting steel. Self tapping screws with machine screw threads may be used in the cabinet assembly. Sheet metal screws with sheet metal threads will not be permitted.

3.1.2 Finishes. - Corrosion resisting steel where exposed in final assembly, except inside of machinery compartment, shall have a number 4 finish as specified in QQ-S-766.

3.1.3 Thermal insulation. - Thermal insulation, other than pipe insulation shall be in accordance with class 5 of MIL-I-16022. Insulation shall be applied in such a manner as to preclude the possibility of settling in service. The insulation shall be installed with sufficient thickness so that condensation will not occur on any external surface when operating under the conditons of 100° F. ambient temperature with relative humidity of not less than 50 percent. The insulation shall be provided with a vapor barrier and designed to prevent accumulation of moisture and possible saturation of the insulation. All insulation spaces shall be filled so as to leave no voids.

3.1.4 Interchangeability. - Similar parts, including repair parts, or corresponding apparatus furnished on the same contract or order or built to the same drawings, shall be strictly interchangeable without the necessity of further machining or hand fitting of any kind.

3.2 General assembly. - Each unit shall be factory assembled, complete, self-contained with the refrigeration equipment dehydrated and charged with the necessary operating quantities of refrigerant and oil. The unit shall be ready for operation after removal of shipping protection, opening of valves and connection to electric power.

3.2.1 Valves, controls, and equipment subject to service and repair shall be readily accessible for servicing. Where panels are provided for access to equipment, machine screws may be used for fastening.

3.2.2 Suitable water supply, drain, overflow and vent connections shall be provided as necessary.

3.3 General design. - The equipment shall be designed for operation under shipboard conditons of vibration, roll, pitch and shock. The refrigerating equipment shall be designed, constructed and assembled in accordance with the publications listed in 2.2, unless otherwise specified herein, or in the contract or order.

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3.3.1 The unit shall have adequate structural strength to withstand the test requirements specified in Specification MIL-S-901 (see 4.5.1).

3.3.2 The equipment shall operate satisfactorily when inclined at an angle 15 degrees each side of the vertical in each of two vertical planes at right angles to each other (see 4.5.3).

3.3.3 The moisture content of the refrigerating system shall be not greater than 15 parts per milllion (p.p.m.) in the liquid phase after being dehydrated and charged with refrigerant.

3.3.4 The assembled unit shall be splash proof when tested under the conditons specifiedn in 4.5.2.

3.4 Condensing unit. - The condensing unit shall be designed for operation with dichlorodifluoromethane (R-12) refrigerant. The unit shall contain a hermetic compressor suitable for the intended application.

3.4.1 Compressor. - The hermetic compressor shall consist of a compressor and motor, enclosed in a gas-tight shell. Provision shall be made for adequate lubrication of all rubbing and wearing surfaces under shipboard conditions of inclination and rolling.

3.4.2 Condenser. - The condenser shall be water-cooled and designed for use with sea water at an entering temperature of 85° F. Water velocities in the condenser shall not exceed 6 feet per second (f.p.s.). All parts in contact with sea water shall be made of 90 - 10 copper-nickel conforming to Specification MIL-T-15005.

3.4.3 Water-regulating valves. - A water-regulating valve shall be provided with each condenser. The valve shall be a modulating type which opens on refrigerant pressure increase. The valve shall be suitable for sea water and shall be used to modulate the flow of water required for the condenser. The valve shall be direct-acting actuated by condenser gas pressure. The valves shall be selected to regulate the water flow from shut-off to required capacity within a maximum operating gas pressure rise of 40 pounds per square inch (p.s.i.). The valve shall be adjustable within an operating range of 90 to 140 p.s.i. refrigerant gas pressure. The capacity of the water-regulating valve shall at least equal the water requirement of the condenser, based on available water pressure drop of 15 p.s.i. The valve shall be suitable for a maximum water operating pressure of 150 pounds per square inch gage (p.s.i.g.).

3.4.3.1 The valve shall be constructed of nonferrous or corrosion-resisting material. The valve body shall be made of valve bronze conforming to Specification MIL-B-16541. In addition, the internal metal parts subject to corrosion or erosion shall be made of nickel-copper conforming to Specification QQ-N-281. The valve may be provided with flanged connectors in accordance with Specification MIL-F-20042 or with union ends in accordance with Specification MIL-F-1183. The valve shall be constructed in such a manner as to prevent the possibility of entry of sea water to the refrigerant system in event of derangement.

3.5 Piping. - All piping necessary for the satisfactory operation of the equipment shall be provided. All fittings and joints, where practical, shall be brazed with silver brazing alloy conforming to grade III or IV of Specification MIL-S-15395 or phosphorus copper in accordance with Specification MIL-C-20158. Flanged or flared connections may be provided on the suction and discharge shut-off valves and accessories to permit possible replacement. Other threaded or screwed joints shall be seal-welded or brazed. Connections shall be kept to a minimum. Piping shall be securely supported to minimize strain and vibration.

3.5.1 Where silver brazing is used on corrosion resistant steel the brazing alloy shall conform to grade V silver solder of Specification MIL-C-15395.

3.5.2 Piping system shall withstand the pressure tests specified in 4.3.1.

3.5.3 Compressor suction piping and cold water piping within machinery compartment shall be insulated to prevent condensation.

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3.5.4 Refrigerant piping. - Refrigerant piping shall conform to Specification WW-T-799 and with wall thickness not less than that required for type N. All fittings shall be wrought or forged copper.

3.5.4.1 A disposable type dehydrator shall be provided in the refrigerant circuit. Each dehydrator shall be equipped with an auxiliary screen or other protective means at the dehydrator outlet to prevent passage of dehydrating agent in the event of rupture of the cartridge screen outlet. The desiccant shall be silica gel or activated alumina or a type satisfactory to the bureau or agency concerned.

3.5.5 Water piping. - Suitable drain and overflow connections from the master tank shall be provided. The piping and fittings shall be made of brass, copper or corrosion resistant steel. The internal piping connection shall be made with silver-solder joints. The external connection shall terminate in the back and be a 3/4 inch iron pipe size (i.p.s.) female threaded fitting. Fittings for external connection shall be secured to the frame or to a bracket secured to a frame in such a manner that when the connection is made to the external drain line no strain will be placed upon the internal piping.

3.5.5.1 The drain shall be provided with a valve. The handle of the valve shall extend to a point where it can be reached from the front without removing any panels or parts.

3.6 Electrical equipment. - Unless otherwise specified herein, the electrical equipment shall conform to Specifications MIL-E-917, MIL-C-2212, and National Electrical Manufacturers Association Publication MGI-1955, part 7, as applicable. No portion of the electrical circuit shall be grounded. The frames or enclosures of all electrical components shall be grounded to the cabinet to eliminate hazard from shorts or grounds within the equipment. Equipment and terminals shall be adequately protected against moisture, dirt, mechanical damage and accidental contact by personnel. Equipment shall be furnished for 115 volts, single-phase, 60 cycle, alternating current power supply.

3.6.1 The hermetic unit shall be protected against overload by a thermal protector built into the motor. The thermal protector shall be either the manual or automatic reset type. A switch conforming to Specification MIL-C-2212 or MIL-S-15291 shall be provided. The switch shall open both sides of the circuit.

3.6.1.1 Under operating conditions specified in 3.14.3 and 3.15.3, the temperature rise of the hermetic motor shall not exceed 70°C.

3.6.1.2 Motor starting relays. - Motor starting relays, used in lieu of centrifugal switches for disconnecting the starting winding of single-phase, a.c. motors, shall be dripproof in accordance with Specification MIL-E-2036. Rubber or other shockmounts which are subject to deterioration in service shall not be used for mounting the relays.

3.6.2 Wiring. - The wiring between the electric devices (motors, motor starters, control devices), shall conform to Underwriters Laboratories Publication UL303. The wiring shall be complete and the equipment shall be ready for connection of the incoming power cable at installation.

3.6.3 Insulation resistance. - The insulation resistance of the electrical circuit shall be not less than 10 megohms (see 4.3.3).

3.6.4 Dielectric strength. - The equipment shall be capable of withstanding for a period of 1 minute a dielectric alternating current, root mean square test voltage of 900 volts (see 4.3.4).

3.7 Automatic refrigeration control devices. - Automatic refrigeration control devices shall be readily accessible for adjustment and servicing and shall be of a type satisfactory to the bureau or agency concerned. Control switches shall be of a type that have been tested and granted qualification approval under Specification MIL-C-2212. The enclosure shall be spraytight.

3.7.1 The automatic operation of the compressor shall be controlled by means of a temperature control switch. The switch shall be adjustable from 60°F. to 75°F. The control shall be set to maintain the water temperature at 68° ± 1°F.

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3.7.2 The unit shall also contain a high pressure switch to protect the compressor in the event of water failure.

3.8 Cabinet. - The cabinet enclosure shall be constructed of corrosion resistant steel not less than 0.037 inches thick. All gas or electric arc welded joints visible in final assembly shall be ground and polished to match the surfaces adjacent to the welds. All edges of sheets visible on the front or sides of cabinet exterior shall be smooth and free from burrs. No wood shall be used in the cabinet construction.

3.8.1 The cabinet and condensing unit sections shall have an integral structural frame. The structural assembly shall be welded in accordance with Standard MIL-STD-278 and treated for corrosion resistance after fabrication. Provision shall be made in the frame to secure the unit to the deck.

3.9 Water tanks. - The master tank liner and insert tanks shall be made of corrosion-resisting steel suitable for the intended application. Connections or joints in the master or insert tanks shall be welded or silver brazed.

3.9.1 Master tanks. - The master tank shall contain bath water and be provided with three removable insert tanks and a refrigerant coil. The master tank shall have an insulated overall lid with clamp for securing the unit when not in use. Provision shall be made in the tank to insert the thermostat bulb.

3.9.1.1 Refrigerant coil. - The cooling coil shall be nonferrous tubing wound around the outside of the the master tank liner and continously bonded to it.

3.9.2 Insert tanks. - The insert tanks shall contain photographic solutions and wash water. The tanks shall be provided with splash baffles, splash proof insert covers and hanger bar.

3.10 Identification plates. - Identification plates and other designating marking plates shall be type A in accordance with Specification MIL-I-15024 and shall be of brass or corrosion resisting steel. The identification plates shall be located on the front.

3.11 Repair parts. - The procedure for furnishing and identifying repair parts shall be in accordance with Specification MIL-R-15137. The number of sets shall be as specified (see 6.1). The repair parts shall include those specified in table I, as applicable, and shall constitute one set.

Table I - Repair parts.

Item
Miscellaneous repair parts:
1 - Valve, of each size and type used
1 - Control switch of each type used
1 - Liquid strainer basket (where used)
1 - Capillary tube assembly (where used)
1 - Dehydrator
2 - Sets of gaskets of each type used
Electric repair parts:
Hermetic compressor motor:
1 - Relay
1 - Overload
1 - Capacitor of each size

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3.12 Drawings. - Drawings delineating the equipment shall be furnished. Separate drawings shall be submitted for each type and size. The drawings shall be in accordance with types I and II of Specification MIL-D-963.

3.12.1 Assembly and component drawings shall be furnished as type I. The drawings shall include outline and mounting dimensions, performance data or performance curves. Drawings shall be provided in sufficient detail to show design, construction, dimensions, and operation or function and identity of the parts. Type I drawings shall not contain reference to the contract or ships concerned.

3.12.2 Equipment index drawings. - Type II drawings shall be contract drawings and shall list all the equipment furnished under the contract or order. The drawings shall include one or more indexes titled "List of Equipment and Certification Data", "Onboard Repair Parts" and "List of Drawings". These may be furnished as separate drawings or combined with the information listed in appropriate tables.

3.12.2.1 The "List of Equipment and Certification Data" drawing shall contain a complete itemization of all components being furnished by the contractor including any explanatory information necessary to supplement the type I drawings referenced thereon and to amplify characteristics of the equipment furnished. This drawing also shall contain a diagrammatic piping arrangement, schematic wiring diagram, a weight table showing the weight of the complete unit, and the total and operating refrigerant charge of the assembly. The "Onboard Repair Parts" drawing shall list the repair parts furnished under the contract. The "List of Drawings" shall index all the component and assembly drawings applicable to the contract or order. The list of drawings may be included on the "Onboard Repair Parts" drawing as a separate tabulation.

3.13 Technical manuals. - Technical manuals furnished shall be type III of Specification MIL-M-15071. A photograph of the equipment, shall be included in the general description. An 8 inch X 10 inch print of the equipment photograph shall be forwarded to the bureau or agency concerned. A section shall be provided containing reduced drawings including copies of all drawings required to amplify or illustrate the text of the manual including "List of Material and Certification Data", "Onboard Repair Parts" and "List of Drawings". The quantity of manuals furnished shall be as specified (see 6.1). Where identical manuals have been distributed to addresses (see 6.1) additional manuals need not be furnished except those manuals packaged with the equipment and the quantity of manuals specified for the Spare Parts Control Center, Mechancisburg, Pennsylvania stock.

3.14 Type I - X-ray film developing tank. - The X-ray film developing tank shall consist of a condensing unit with water cooled condenser, water cooling unit, controls, accessories and three insert tanks suitable for developing film, with maximum dimensions of 14 inches X 17 inches all supported and enclosed in a metal cabinet.

3.14.1 Size 1 - Cabinet. - The cabinet shall not be greater than 24 inches wide by 22 inches deep by 45 inches high exclusive of lid or hardware.

3.14.1.1 One insert tank shall have a capacity of at least 5 gallons. The other two insert tanks shall have a capacity of at least 3 gallons each.

3.14.2 Size 2 - Cabinet. - The cabinet shall not be greater than 34 inches wide X 22 inches deep by 45 inches high exclusive of lid or hardware.

3.14.2.1 One insert tank shall have a capacity of at least 10 gallons. The other two insert tanks shall have a capacity of at least 5 gallons each.

3.14.3 Operating requirement. - The unit shall automatically produce and maintain an average process solution temperature in the master tank of 68° F. ± 1° F. at an ambient temperature of 110° F. The mechanical cooling apparatus shall be capable of reducing the required quantity of water and developing solutions in the insert tanks from a temperature of 100° F. to 68° F. in approximately 4 hours or less.

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3.15 Type II - Photodosimetric film developing tank. - The photodosimetric film developing tank shall consist of a condensing unit with water cooled condenser, water cooling unit, water heating unit, controls accessories and three insert tanks all supported and enclosed in a metal cabinet.

3.15.1 The photodosimetric unit shall contain an electrical heating element to heat the bath water in the event the water is below 68° F. Selection of heating or cooling only shall be by a suitable manually controlled selector switch.

3.15.2 The cabinet shall not be greater than 18 inches wide by 17 inches deep by 36 inches maximum height exclusive of lid or hardware.

3.15.2.1 Each insert tank shall have a capacity of at least 2 gallons.

3.15.3 Operating requirement. - The unit shall automatically produce and maintain an average process solution temperature in the master tank of 68° F ± 1° F. at an ambient temperature of 110° F. The mechanical cooling apparatus shall be capable of reducing the required quantity of water and developing solutions in the insert tanks from a temperature of 100° F. to 68° F. in approximately 4 hours or less.

3.15.3.1 The heating element shall be capable of heating the water and developing solutions in the insert tanks from a temperature of 50° F. to 68° F. in approximately 2 hours or less in an ambient of 50° F.

3.16 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issured, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling for inspection. -

4.2.1 Lot. - All refrigerating equipment of the same type produced at one time shall be considered a lot for purposes of inspection.

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4.2.2 From each production lot the inspector shall select sample units in accordance with table II and inspect each sample unit for adjustment, fits, leaks, material, finish, and general conformance with the requirements of this specification. Each sample unit shall be subjected to the inspection specified in 4.4. If any unit is found nonconforming in any requirement it shall be counted a defective unit and the lot shall be rejected. A rejected lot may be offered again for inspection after the manufacturer has removed all defective units or corrected the deficiency.

Table II - Sampling for inspection.

Lot size number of units	Sample size number of units
1 to 5	All
6 to 8	6
9 to 15	8
16 to 25	10
26 to 40	13
41 to 65	17

4.2.3 The tests of 4.3.1 to 4.3.5 shall be repeated on each sample unit selected in accordance with 4.2.2 and the results compared with the manufacturer's records. If any discripancies are found, all units shall be re-tested to verify the records.

4.3 Production tests. - The following tests shall be performed by the manufacturer on each unit.

4.3.1 Pressure. - All component parts, except compressor unit and controls, on the high-pressure side of the refrigeration equipment shall withstand without damage an air or gas test of 300 p.s.i. The component parts, except compressor unit and controls, on the low-pressure side of the refrigerating system shall withstand without damage an air or gas test of 250 p.s.i.

4.3.2 Leakage test. - The refrigerating system shall be leak tested by means of a halide leak detector. The unit shall be free of evidence of leaks when the system is subjected to a minimum gas pressure of 235 p.s.i.g. on the high side and a minimum gas pressure of 140 p.s.i.g. on the low side.

4.3.3 Insulation resistance test. - With the motor and electrical circuit at ambient temperature the insulation resistance shall be as specified in 3.6.3. The measuring of insulation shall be made by connecting the two live terminals together and measuring the resistance between terminals and ground. The insulation resistance shall be measured with a 500 volt insulation-resistance meter conforming to type GC of Specification MIL-O-16485. The time of test voltage application shall be not less than 60 seconds.

4.3.4 Dielectric strength test. - With the motor and electrical circuit at ambient temperature the dielectric strength shall be as specified in 3.6.4. The dielectric strength test shall be conducted with the line terminals together.

4.3.5 Operating tests. - The units shall be operated for a period of at least one hour with controls set for specified operating conditons. At the conclusion of this operational test, the entire refrigerant circuit connections under refrigerant pressure shall be tested with a halide leak detector to determine leakage. In the event of leakage, all leaks shall be repaired and the performance test shall be repeated. In addition it shall be determined that the water containing parts are tight. The heating element in the photodosimetric unit shall be operated long enough to verify that it operates satisfactorily.

4.3.5.1 During this test it shall be verified that the controls are adjusted and functioning properly.

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4.3.5.2 Under equivalent ambient temperature, the electrical power imput shall be measured and compared with the input of all other units which have been tested. If any unit required 7 percent more power than the average of all the acceptable units, it shall be rejected until the cause has been found and corrected.

4.4 Acceptance inspection. - Each sample unit selected as specified in 4.2.2 shall be inspected for adjustment, fits, leaks, material, finish and general conformance with this specification. The inspector shall test the units to determine compliance with the following:

(a) The compressor valve seal caps and all other seal caps in the refrigerant system shall be provided with a gasket, shall be tight for shipment and shall be tight against refrigerant leaks.
(b) Bolts, nuts, and screws shall be tight; equipment and parts shall be properly fastened and secured.
(c) No parts shall be fractured, split, torn, dented or otherwise damaged such as to affect serviceability.
(d) There shall be no sharp or ragged edges on the sheeting that may be injurious to personnel.
(e) All cold lines shall be properly insulated (see 3.5.3).
(f) The disconnect switch shall operate satisfactorily (see 3.6.1).
(g) The insulation resistance to ground shall be satisfactory (see 3.6.3 and 4.3.3).
(h) Dielectric strength test shall be satisfactory (see 3.6.4 and 4.3.4).
(i) The temperature controls shall be properly set and function (see 3.7.1 and 4.3.5.1).
(j) The unit including limiting and mounting dimensions shall be in accordance with approved drawings (see 3.12).
(k) The technical manuals shall be packed with the unit.
(ℓ) The refrigerant system shall be free of refrigerant leaks (see 4.3.5).
(m) The power consumption for any unit shall not be excessive (see 4.3.5.2).
(n) The overall finish of the unit shall be such that it does not detract from its overall appearance.

4.5 Acceptance tests. - Acceptance tests on a unit of each equipment type and design shall be conducted as applicable and a test report submitted to the bureau or agency concerned. Tests shall be conducted as required in 4.5.1 to 4.5.6.1 inclusive. The unit that has satisfactorily passed the tests may be used as one of the production units after correction of any defects. The contractor may acquire all materials required for use on the contract prior to approval of the unit being tested. Any final assembly work performed by the contractor prior to approval of the unit shall be at the contractor's risk. Where acceptances tests as required by this specification have previously been conducted and equipment offered is the same as that previously tested and accepted a test report of prior tests which has been accepted by the bureau or agency concerned shall be acceptable in lieu of retesting. The acceptance tests with the exception of the structural strength test shall be conducted at the manufacturers' plant in the presence of a government inspector.

4.5.1 Structural strength test. - The unit shall be tested as required by Specification MIL-S-901. The test shall be conducted at a laboratory designated by the bureau or agency concerned (see 6.1). All expenses incurred in shipping the unit to and from the designated laboratory shall be borne by the contractor. The costs of conducting the structural strength test shall be borne by the government. Failure to operate satisfactorily or evidence of breakage or disassembly at the conclusion of this test shall be cause for rejection.

4.5.2 Splash proof provisions. - Water shall not be splashed into or out of the insert tanks when the tank and insert covers are in place and the axis is rocked to a tilt of 60 degrees back and forth and side to side (in each case a total excursion of 120 degrees) five times. Each vibration to be made in about one second.

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4.5.3 Inclination test. - The unit with covers removed shall be inclined at an angle of 15 degrees each side of the vertical planes at right angles to each other and operated at least one hour in each plane with no abnormal variations in temperature.

4.5.4 Design capacity test. - Observations shall be taken at not less than 20 minute intervals. Ambient temperature, temperature of water in master tank, voltage at service connections, refrigerant suction and discharge pressures or temperatures shall be taken. The unit shall be installed in an ambient temperature of 110°F., condensing water maintained at 85°F. and controls set for automatic operation. The test shall be started with water in the master and insert tanks of 100°F. and continued until the water in the insert tanks has been reduced to 68°F.. The water temperature in the insert tanks shall be reduced to 68°F. in approximately 4 hours or less.

4.5.4.1 The temperature rise of the compressor motor winding shall be determined at the end of the test period. The temperature rise shall not be greater than that specified in 3.6.1.1. The temperature rise of the running winding in the compressor-motor shall be measured and computed by the resistance method.

4.5.5 Standby operation test. - This test shall start with an ambient temperature of 110°F., water in the master and insert tanks of 68°F. and the controls set for automatic operation. The temperature of the water in the insert tanks shall be maintained at 68°F ± 1°F. for a period of 2 hours.

4.5.6 Heating test (type II). - The unit shall be installed in an ambient temperature of 50°F. with controls set for automatic operation. The heating unit shall be started with water in the master and insert tanks at 50°F. and continued until the water in the insert tanks has been heated to 68°F.. The water temperature in the insert tanks shall be heated to 68°F. in approximately 2 hours or less.

4.5.6.1 Electrical insulation resistance (type II). - The insulation resistance between heater coil and ground shall equal or exceed 100 megohms cold and one megohm hot.

4.5.7 A copy of the test report and inspection report should be submitted to the bureau or agency concerned.

4.6 Inspection of preparation for delivery. - Sample units and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with the requirements of section 5 herein.

4.7 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Refrigeration equipment shall be preserved and packaged in accordance with Specification MIL-P-12323, as applicable, except as otherwise specified herein.

5.1.1.1 Extent of disassembly shall be determined by the contractor, subject to the approval of the Government inspector.

5.1.1.2 Strainers and dehydrators shall not be coated with preservative compound.

5.1.1.3 Repair parts. -

5.1.1.3.1 Electrical. - Electrical repair parts shall be preserved and packaged in accordance with Specification MIL-P-17555.

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5.1.1.3.2 Valves. - Valves and fittings, except refrigerant valves, shall be preserved and packaged in accordance with Specification MIL-P-3. Refrigerant valves shall be individually packaged in accordance with Method IA of Specification MIL-P-116. Contact preservatives will not be required for refrigerant valves.

5.1.1.3.3 Miscellaneous repair parts. -

5.1.1.3.3.1 Gasket sets. - Gaskets shall be individually placed between sheets of paperboard or fiberboard of sufficient stiffness to resist bending. Each set shall then be packaged in accordance with Method IA of Specification MIL-P-116.

5.1.1.3.3.2 Strainer basket, dehydrator and capillary tube assembly shall be individually packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 Level C. - The equipment shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. -

5.2.1 Level A and B. -

5.2.1.1 Equipment. - The equipment shall be packed for overseas or domestic shipment as specified (see 6.1) in accordance with Specification MIL-P-12323.

5.2.1.2 Repair parts shall be packed for the level specified (see 6.1) in accordance with Specification MIL-P-17555.

5.2.2 Level C. - Equipment and repair parts shall be packed in containers in a manner which will insure acceptance by common carrier and safe delivery at destination. Containers or method of shipment shall comply to the carrier's rules and requlations applicable to the mode of transportation.

5.3 Technical publications. - Technical manuals, pamphlets or handbooks, as applicable shall be packaged, packed and marked for the level specified (see 6.1) in accordance with Specification MIL-M-15071.

5.4 Drawings. - Drawings shall be packaged, packed and marked for the level specified (see 6.1) in accordance with Specification MIL-D-963.

5.5 Marking. - In addition to any special marking required in the contact or order, interior packages and shipping container shall be marked in accordance with Specification MIL-P-12323 and Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size and quantity (see 1.2, 3.14 and 3.15).
(c) Number of sets of repair parts required (see 3.11).
(d) Number of technical manuals required (see 3.13).
    (1) The following should be furnished:
        a. 1 copy with each unit.
        b. 2 copies for the Government inspector for inspection purposes.
        c. 3 copies for the bureau or agency concerned.
        d. 2 copies for each U.S. Naval Shipyard.
        e. 2 copies for each unit furnished for stock.
(e) Laboratory where tests will be conducted (see 4.5.1).
(f) Selection of applicable levels of preservation, packaging and packing (see 5.1, 5.2, 5.3 and 5.4).

6.1.1 Bid data. - Bidders shall submit, with their bids, a statement in duplicate showing:

(a) Itemized list and total weight of the equipment, including repair parts. (Repair parts to be separately priced.)
(b) Compressor capacity, bore or cylinders and stroke, revolutions per minute, and displacement.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 6635-0003Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-T-45366E

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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NAVY DEPARTMENT

OFFICIAL BUSINESS

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WASHINGTON, D.C. 20360

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